The Logical Cost of Fermat's Last Theorem: A Constructive Reverse Mathematics Audit of Wiles's Proof

(Paper 68, Constructive Reverse Mathematics Series)

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Abstract

We perform a stage-by-stage constructive reverse mathematics audit of Wiles's proof that the p-adic Galois representation $\rho_{E,p}$ attached to every semistable elliptic curve $E/\mathbb{Q}$ is modular, and hence of Fermat's Last Theorem. An important distinction: this audit concerns *Galois modularity* ($\rho_{E,p} \cong \rho_{f,p}$ for some eigenform f), which is what Ribet's level-lowering theorem requires. *Geometric modularity* (that E is isogenous to the abelian variety A_f) additionally requires Faltings's Isogeny Theorem, whose Archimedean content places it at CLASS; FLT does not need it.

The proof decomposes into five stages: residual modularity (Langlands–Tunnell), deformation ring construction, Hecke algebra construction, the numerical criterion, and Taylor–Wiles patching. Our principal finding is an *asymmetry*: Stages 2–5 are fully constructive (BISH), while Stage 1 requires WLPO (Weak Limited Principle of Omniscience). The non-constructive content of Wiles's proof is concentrated entirely at its entry point—the analytic theory of weight 1 automorphic forms—and the Taylor–Wiles engine contributes zero logical cost.

Two post-Wiles developments drive the Stage 5 classification: Brochard's proof of de Smit's conjecture (2017), which eliminates the infinite inverse limit, and unconditional effective Chebotarev bounds (Lagarias–Montgomery–Odlyzko 1979, Ahn–Kwon 2019), which make the Taylor–Wiles prime search a bounded computation. Neither result was motivated by constructive foundations; the community unknowingly eliminated the Fan Theorem from the proof of FLT over twenty-two years.

The Lean 4 verification (493 lines across three files, zero `sorry`) formalizes the logical assembly: deep theorems are axiomatized and flagged; the conditional classification is machine-checked.

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*Lean 4 formalization available at https://doi.org/10.5281/zenodo.18748460.

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1 Introduction

1.1 Main results

What is the logical cost of proving Fermat's Last Theorem? The question is not about computational complexity but about logical *principles*: which axioms beyond intuitionistic logic are

needed? This paper answers the question for Wiles's proof [19], as refined by Taylor–Wiles [17], Diamond [10], and Brochard [5]. Throughout, "modularity" means *Galois modularity*: the p-adic representation $\rho_{E,p}$ is isomorphic to the representation attached to a weight 2 eigenform. Ribet's level-lowering theorem operates on $\bar{\rho}_{E,p}$ and yields the FLT contradiction $S_2(\Gamma_0(2)) = 0$ without ever requiring the geometric statement that E is isogenous to A_f. Geometric modularity requires Faltings's Isogeny Theorem (Archimedean Green's functions, integration over $E(\mathbb{C})$), placing it at CLASS; it plays no role in FLT. We establish four results:

Theorem A (Stage 5 is BISH). The Taylor–Wiles patching argument, in the Brochard formulation, is a BISH-decidable finite computation. Brochard's finite-level criterion [5] eliminates the Fan Theorem; effective Chebotarev bounds [13, 1] eliminate Markov's Principle. The logical cost descends from $\mathsf{MP} + \mathsf{FT}$ (1995) to BISH (2017/2019).

Theorem B (Stages 2–4 are BISH). The deformation ring (Schlessinger, Fontaine–Laffaille), Hecke algebra (finite arithmetic), numerical criterion (subsumed by patching), and CM base case (Rubin's Euler system with effective Chebotarev) are all BISH.

Theorem C (Stage 1 requires WLPO). The Langlands–Tunnell theorem requires at least WLPO: the Arthur–Selberg trace formula involves spectral decomposition of L^2-spaces (eigenvalue isolation: WLPO), orbital integral matching (real equality: WLPO), and the converse theorem for GL_3 (Phragmén–Lindelöf condition).

Theorem D (Asymmetry Theorem). The overall classification of Wiles's proof is $\mathsf{BISH} + \mathsf{WLPO}$. The WLPO is localized entirely in Stage 1. Removing Stage 1 drops the classification to BISH.

1.2 Constructive Reverse Mathematics: a brief primer

CRM calibrates mathematical statements against logical principles of increasing strength within Bishop-style constructive mathematics (BISH). The hierarchy relevant to this paper is:

$$\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{MP} \subset \mathsf{BISH} + \mathsf{LLPO} \subset \mathsf{BISH} + \mathsf{WLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}.$$

Here WLPO (Weak Limited Principle of Omniscience) states that every binary sequence is identically zero or not; equivalently, $\forall x \in \mathbb{R},\ x = 0 \vee \neg(x = 0)$. MP (Markov's Principle) states that a binary sequence that is not all zeros contains a 1; it converts $\neg\neg\exists$ to $\exists$. FT (Fan Theorem) is the constructive form of König's Lemma. For a thorough treatment of CRM, see Bridges–Richman [4]; for the broader program of which this paper is part, see Papers 1–67 of this series and the atlas survey [22].

1.3 Current state of the art

McLarty [15] asked what *set-theoretic* strength FLT requires, showing that Grothendieck universes are eliminable. Our question is orthogonal: we ask what *constructive* principles the proof requires. McLarty's analysis concerns the ambient set theory; ours concerns the logical content of the mathematical arguments.

No prior work has applied constructive reverse mathematics to the logical structure of Wiles's proof or the Taylor–Wiles method. The constructive calibration is novel.

1.4 Position in the atlas

The present paper is part of the Constructive Reverse Mathematics program (Papers 1–67). Paper 67 [24] synthesizes the arithmetic geometry phase (Papers 45–66); Paper 40 [21] covers the physics phase; Paper 50 [22] provides the atlas framework. Paper 59 [23] classified the p-adic comparison (Fontaine–Laffaille) as BISH, directly supporting Stage 2.

Paper 68 opens a new direction: classifying *proof methods* rather than theorems. The de-omniscientizing descent pattern identified in Paper 50 for the five great conjectures of arithmetic geometry reappears here in the *evolution* of the Taylor–Wiles method over 1995–2017.

2 Preliminaries

Definition 2.1 (Weak Limited Principle of Omniscience)**.** WLPO: For every binary sequence $\alpha : \mathbb{N} \to \{0, 1\}$, either $\forall n,\ \alpha(n) = 0$ or $\neg(\forall n,\ \alpha(n) = 0)$. Equivalently, for every $x \in \mathbb{R}$: $x = 0 \vee \neg(x = 0)$.

Definition 2.2 (Markov's Principle)**.** MP asserts: for every binary sequence $\alpha\colon \mathbb{N} \to \{0, 1\}$,

$$\neg(\forall n,\ \alpha(n) = 0) \implies \exists n,\ \alpha(n) = 1.$$

This converts double-negated existence to constructive existence.

Definition 2.3 (Fan Theorem)**.** FT: Every decidable bar of the full binary fan (Cantor space) is uniform. Equivalently, Cantor space $2^{\mathbb{N}}$ is compact. Over BISH, FT is equivalent to WKL_0 (every infinite binary tree has an infinite path).

Definition 2.4 (The five stages of Wiles's proof)**.** Let $E/\mathbb{Q}$ be a semistable elliptic curve with conductor N, and let $p = 3$. Let $\rho : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{Z}_3)$ be the 3-adic Galois representation on the Tate module $T_3(E)$, and $\bar{\rho} : G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F}_3)$ its reduction. Following the standard decomposition [8, 9], Wiles's proof that ρ is modular proceeds in five stages:

Stage 1 (Residual modularity): prove $\bar{\rho}$ is modular via Langlands–Tunnell [12, 18].
Stage 2 (Deformation ring): construct the universal deformation ring R.
Stage 3 (Hecke algebra): construct $\mathbb{T}$ localized at $\mathfrak{m}$.
Stage 4 (Numerical criterion): verify the Wiles–Lenstra numerical criterion.
Stage 5 (Patching): select Taylor–Wiles primes and prove $R \cong T$.

Definition 2.5 (Taylor–Wiles primes)**.** A set $Q = \{q_1, \ldots, q_r\}$ of primes are *Taylor–Wiles primes at level n* if for each $q \in Q$: (i) $q \equiv 1 \pmod{p^n}$, (ii) $q \nmid N$, (iii) $\bar{\rho}(\mathrm{Frob}_q)$ has distinct eigenvalues in $\bar{\mathbb{F}}_p$.

Remark 2.6 (Historical note on Stage 4)*.* In the published 1995 proof [19], Stage 4 is subsumed by Stage 5: the numerical criterion is verified *within* the patching argument, not by an external Euler system. The only Euler system remaining is Rubin's [16], used for the CM base case.

3 Main Results

3.1 Theorem A: Stage 5 is BISH

We classify Stage 5 first because the result is the most surprising: the heart of the Taylor–Wiles method is fully constructive.

3.1.1 The original argument and its logical cost

Taylor–Wiles [17] and Diamond [10] select TW primes at each level $n \geq 1$, form the projective system $\{R_{Q_n}/\mathfrak{m}^n\}_{n\geq 1}$, and take the inverse limit. The nonemptiness of the inverse limit of nonempty finite sets is a compactness argument equivalent to WKL_0. Diamond [10] abstracts the freeness criterion but *still* requires the inverse limit. The logical cost is $\mathsf{BISH} + \mathsf{MP} + \mathsf{FT}$: MP from the unbounded search for TW primes at each level n, and FT from the inverse limit compactness.

3.1.2 Brochard's elimination of the Fan Theorem

Theorem 3.1 (Brochard 2017, Theorem 1.1)**.** *Let $A \to B$ be a local morphism of commutative Artinian local rings. If $\operatorname{edim}(B) \leq \operatorname{edim}(A)$ and M is a nonzero B-module that is flat as an A-module, then M is free as a B-module and B is a relative complete intersection over A.*

Corollary 3.2 (Elimination of patching)**.** *The entire Taylor–Wiles patching argument can be performed at a single finite level ($n = 2$), without forming any infinite projective system.*

The logical cost of Stage 5 compactness drops from FT to BISH: the infinite inverse limit is replaced by a finite-level algebraic check.

3.1.3 Effective Chebotarev eliminates Markov's Principle

Theorem 3.3 (Effective Chebotarev)**.** *Let $L/\mathbb{Q}$ be a finite Galois extension with absolute discriminant d_L, and let C be a conjugacy class in $\operatorname{Gal}(L/\mathbb{Q})$. Unconditionally, there exists a prime $q \leq d_L^{12577}$ with $\operatorname{Frob}_q \in C$ (Lagarias–Montgomery–Odlyzko [13], Ahn–Kwon [1]).*

Proposition 3.4 (TW prime search is BISH)**.** *The search for Taylor–Wiles primes at level $n = 2$ is a BISH-decidable finite computation: compute d_{L_2} from $(N, p, \bar{\rho})$, then test all primes $q \leq d_{L_2}^{12577}$ for conditions (i)–(iii). The bound is astronomically large but computable before the search begins.*

Proof. The splitting field L_2 is the compositum of the splitting field $K_{\bar{\rho}}$ of $\bar{\rho}$ and $\mathbb{Q}(\mu_{p^2})$. The absolute discriminant d_{L_2} is computable from $(N, p, \bar{\rho})$ by standard algebraic number theory (Hensel bounds for wild ramification). By Theorem 3.3, there exists a prime $q \leq d_{L_2}^{12577}$ with Frob_q in the appropriate conjugacy class. The TW conditions (i)–(iii) are decidable for each prime (finite arithmetic). The search terminates within $d_{L_2}^{12577}$ steps. □

Theorem 3.5 (Stage 5 Classification)**.** *Stage 5 of Wiles's proof, in the Brochard formulation, is BISH. The compactness (FT) is eliminated by Theorem 3.1; the prime search (MP) is eliminated by Proposition 3.4.*

Proof. Given axioms B1 (Brochard's criterion), B2 (effective Chebotarev), and B3 (discriminant computability), Stage 5 proceeds as: (1) compute the Chebotarev bound from $(N, p, \bar{\rho})$; (2) search all primes up to the bound for TW conditions; (3) construct patching data at level $n = 2$; (4) apply Brochard's criterion to obtain freeness ($R \cong T$). All steps are finite, decidable computations. No Fan Theorem (infinite inverse limit). No Markov's Principle (unbounded search). □

3.1.4 The historical de-omniscientizing descent

Formulation	Year	Logical cost
Taylor–Wiles (original)	1995	MP + FT
Diamond (algebraic patching)	1997	MP + FT
Brochard (de Smit's conjecture)	2017	MP
Brochard + effective Chebotarev	2017/2019	BISH

None of these refinements were motivated by constructive foundations. The community eliminated the Fan Theorem and Markov's Principle from the proof of FLT without knowing it.

3.2 Theorem B: Stages 2–4 are BISH

3.2.1 Stage 2: The deformation ring (BISH)

The universal deformation ring R is a quotient of $W(\mathbb{F}_p)[[x_1, \ldots, x_d]]$ by an ideal specified by local conditions. The local condition at p (crystalline with Hodge–Tate weights $(0, 1)$) is classified by Fontaine–Laffaille theory; Paper 59 [23] classified the p-adic comparison as BISH. The local conditions at primes $\ell \mid N$ are finite-dimensional specifications over $\mathbb{F}_p$. The universal property (Schlessinger's criterion) involves tangent space and obstruction calculations, both finite algebra. The tangent space $H^1(G_{\mathbb{Q},S}, \mathrm{ad}^0\bar{\rho})$ is finite-dimensional over $\mathbb{F}_p$ (promotion from hull to universal ring follows); all steps are constructive commutative algebra in the sense of [14]. Stage 2 is BISH.

3.2.2 Stage 3: The Hecke algebra (BISH)

The Hecke algebra $T = \mathbb{T}(N, 2) \otimes \mathbb{Z}_p$, localized at $\mathfrak{m}$, is a finitely generated $\mathbb{Z}_p$-algebra. Hecke operators T_ℓ for $\ell \nmid Np$ are explicit linear maps on a finite-dimensional space; diamond operators $\langle d \rangle$ for $d \in (\mathbb{Z}/N\mathbb{Z})^\times$ are explicit; the dimension of $S_2(\Gamma_0(N))$ is computable by Riemann–Roch. Everything is finite arithmetic. Stage 3 is BISH.

3.2.3 Stage 4: The numerical criterion (BISH)

The numerical criterion in the published proof is verified within Stage 5. For the CM base case, Rubin's Euler system [16] selects Kolyvagin primes via effective Chebotarev (Theorem 3.3).

The Euler system machinery beyond the prime search requires audit. Rubin's argument constructs norm-compatible systems of elliptic units $c_n \in K_n^\times$ and applies Kolyvagin's descent. Each step operates within a *finite* extension tower K_n/K; the elliptic units are explicit algebraic numbers; and the descent argument is a finite computation in Galois cohomology $H^1(\mathrm{Gal}(K_n/K), E[p^n])$ with finite coefficients. No topological compactness or unbounded search enters beyond the Kolyvagin prime selection, which is bounded by effective Chebotarev. The Euler system machinery is therefore BISH, and the full CM base case is BISH.

We flag this as an *axiomatized* classification: the Lean formalization records the Stage 4 classification as a definitional assignment. A fully formal audit of Rubin's Euler system would require formalizing the norm-compatibility relations and Kolyvagin's descent in Lean, which is beyond the scope of this paper.

3.2.4 Galois cohomology and Selmer groups (BISH)

Two potential non-constructive sites require explicit treatment.

Decomposition groups. The restriction maps $\mathrm{res}_v : H^1(G_{\mathbb{Q},S}, M) \to H^1(G_{\mathbb{Q}_v}, M)$ require fixing a decomposition group $D_v \subset G_{\mathbb{Q}}$. Classically, this requires $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_v$ (Krull/Zorn). But Wiles works with finite coefficient modules: $\mathrm{ad}^0\bar{\rho}$ is a finite $\mathbb{F}_p$-vector space. Galois cohomology factors through a finite quotient $\mathrm{Gal}(K/\mathbb{Q})$, and the decomposition group is determined by a finite search. No Krull/Zorn is needed. This is BISH.

Poitou–Tate duality. For finite coefficient modules, Poitou–Tate reduces to a long exact sequence of finite abelian groups. The duality is linear algebra over $\mathbb{F}_p$. No topological compactness is needed. This is BISH.

3.3 Theorem C: Stage 1 requires WLPO

This is where the non-constructive content lives.

3.3.1 The weight 1 obstruction

The Langlands–Tunnell theorem asserts that every 2-dimensional complex representation of $G_{\mathbb{Q}}$ with solvable image arises from a weight 1 modular form. Unlike weight $k \geq 2$ forms, weight 1 forms do not appear in the étale cohomology of modular curves. The only known proof route uses the Arthur–Selberg trace formula and analytic properties of automorphic L-functions.

3.3.2 The trace formula (WLPO)

Langlands's proof of cyclic base change [12] uses the Arthur–Selberg trace formula. Three constructive obstructions arise:

Spectral decomposition. Extracting a discrete automorphic representation from the spectral decomposition of $L^2(\mathrm{GL}_2(F)\backslash\mathrm{GL}_2(\mathbb{A}_F))$ requires isolating an eigenvalue from the continuous spectrum. Constructively, this requires deciding whether the spectral measure of an interval is zero or positive—at minimum WLPO.

Orbital integral matching. The Fundamental Lemma (for GL_2, elementary) asserts equality of certain Archimedean orbital integrals over $\mathbb{R}$ and $\mathbb{C}$. Verifying exact equality of real integrals requires WLPO.

Continuous spectrum cancellation. The trace formula isolates the discrete spectrum by subtracting the Eisenstein contribution. Testing whether a specific complex expression equals zero requires WLPO.

3.3.3 The converse theorem (WLPO/LPO)

Tunnell's proof [18] uses the Gelbart–Jacquet symmetric square lifting [11], which relies on the converse theorem for GL_3. The Phragmén–Lindelöf condition (boundedness in vertical strips) requires evaluating a supremum over an unbounded, non-compact domain. This is at least LPO.

Theorem 3.6 (Stage 1 requires WLPO)**.** *The Langlands–Tunnell theorem, as used in Wiles's proof, requires at least* WLPO*. The Phragmén–Lindelöf condition requires evaluating a supremum over an unbounded domain (*LPO*-level in general). For* GL_2*, the Ramanujan conjecture is known (Deligne 1974), providing a computable bound* $|\lambda_\pi(p)| \leq 2p^{(k-1)/2}$ *on the Hecke eigenvalues. This converts the unbounded supremum to a comparison against an explicit threshold: is a specific computable*

quantity zero or not? The reduction from LPO *to* WLPO *follows because a bounded quantity with a computable bound requires only a zero test (*WLPO*), not an unbounded search (*LPO*).*

Remark 3.7 (Open question: WLPO vs. LPO). The precise classification of Stage 1 within the range [WLPO, LPO] depends on whether the spectral gap of $L^2(\mathrm{GL}_2(F)\backslash\mathrm{GL}_2(\mathbb{A}_F))$ is computable. For GL_2, the Ramanujan conjecture is known (Deligne 1974), and the spectral gap should be effective. We state the classification as WLPO with this caveat.

3.4 Theorem D: The Asymmetry Theorem

Theorem 3.8 (Asymmetry of Wiles's Proof)**.** *The* CRM *classification of Wiles's proof is:*

Stage	***Content***	***Classification***	***Key input***
1	*Langlands–Tunnell*	WLPO	*Trace formula, converse theorem*
2	*Deformation ring*	BISH	*Schlessinger, Fontaine–Laffaille*
3	*Hecke algebra*	BISH	*Finite algebra, Riemann–Roch*
4	*Numerical criterion*	BISH	*Subsumed by Stage 5*
4′	*CM base case (Rubin)*	BISH	*Effective Chebotarev*
5	*Patching*	BISH	*Brochard + effective Chebotarev*
Overall		BISH + WLPO	

The WLPO *is localized entirely in Stage 1. Removing Stage 1 drops the classification to* BISH*.*

Proof. Stages 2–5 are classified individually in §3.1 and §3.2: each is BISH. The join (maximum) over all stages is determined by Stage 1, which requires WLPO (Theorem 3.6). Since BISH is subsumed by BISH + WLPO, the overall classification is BISH + WLPO.

For the localization statement: the only axiom beyond BISH used anywhere in the proof is the WLPO content of the trace formula and converse theorem in Stage 1. The BISH stages do not invoke Stage 1's output except as a single boolean datum ("$\bar{\rho}$ is modular: yes/no"), which is a decidable proposition once Stage 1 has been executed. □

Corollary 3.9 (Logical cost of FLT)**.** *Wiles's proof of Fermat's Last Theorem has logical cost* BISH + WLPO*.*

Corollary 3.10 (The engine is constructive)**.** *The Taylor–Wiles patching method—the central proof technology of the Langlands program for* $\mathrm{GL}_2/\mathbb{Q}$*—contributes zero logical cost beyond* BISH*.*

Corollary 3.11 (Algebraic weight 1 modularity implies constructive FLT)**.** *If a purely algebraic proof is found that 2-dimensional Artin representations of solvable type are modular—bypassing the Arthur–Selberg trace formula—then Wiles's proof of FLT becomes fully constructive (*BISH*).*

Remark 3.12 (Galois modularity vs. geometric modularity). Wiles's proof establishes that $\rho_{E,p} \cong \rho_{f,p}$ for a weight 2 eigenform f—*Galois modularity.* The stronger statement that E is isogenous to the abelian variety A_f (*geometric modularity*) requires Faltings's Isogeny Theorem, which evaluates Archimedean Green's functions and integrates over $E(\mathbb{C})$, incurring irreducible CLASS cost. Since Ribet's level-lowering theorem [8] operates entirely on the Galois representation $\bar{\rho}_{E,p}$, the FLT contradiction $S_2(\Gamma_0(2)) = 0$ follows from Galois modularity alone, completely bypassing Faltings. The BISH + WLPO classification of Theorem D applies to Galois modularity and FLT; geometric modularity remains CLASS.

4 CRM Audit

4.1 Constructive strength classification

Result	Strength	Necessary?	Sufficient?
Theorem A (Stage 5 is BISH)	BISH	Yes (from axioms)	Yes
Theorem B (Stages 2–4 are BISH)	BISH	Yes (from axioms)	Yes
Theorem C (Stage 1 requires WLPO)	WLPO	WLPO necessary	WLPO sufficient
Theorem D (Asymmetry)	BISH + WLPO	Yes (join)	Yes

Note on BISH *classification.* Lean's $\mathbb{R}$ and $\mathbb{C}$ (Cauchy completions) pervasively introduce `Classical.choice` as an infrastructure artifact. Constructive stratification is established by *proof content*—explicit witnesses, no omniscience principles as hypotheses—not by axiom checker output (cf. Paper 10, §Methodology).

4.2 What descends, from where, to where

The central CRM phenomenon is a *descent in logical strength* of the patching step:

$$\underbrace{\mathsf{MP} + \mathsf{FT}}_{\text{1995: TW original}} \xrightarrow{\text{Brochard + eff. Chebotarev}} \underbrace{\mathsf{BISH}}_{\text{2017: no omniscience}} .$$

The mechanism: Brochard's finite-level criterion replaces the infinite inverse limit (eliminating FT), and effective Chebotarev bounds the prime search (eliminating MP).

Paper 50 [22] identified the de-omniscientizing descent pattern in the five great conjectures of arithmetic geometry. Paper 68 reveals a parallel phenomenon in *proof methods*: the community de-omniscientized the proof of FLT without knowing it.

4.3 Comparison with earlier calibration patterns

This paper establishes a variant of the pattern from Papers 2, 7, and 8:

1. Identify the constructive obstruction (WLPO for Stage 1; MP + FT for Stage 5 pre-Brochard).
2. Classify each stage independently.
3. Identify a structural bypass (Brochard + effective Chebotarev → BISH for Stage 5).
4. Show the bypass occurred historically without constructive motivation.

The novelty is that the de-omniscientizing descent occurred on a *human* timescale (1995–2017) in the published literature, rather than on a conjectural timescale (the motives conjectures).

5 Formal Verification

5.1 File structure and build status

The Lean 4 bundle resides at `paper 68/P68_WilesFLT/` with the following structure:

File	Lines	Content
`Paper68_Axioms.lean`	132	Opaque types, axioms B1–B3, CRM hierarchy
`Paper68_Stage5.lean`	178	Target 1: Stage 5 is BISH
`Paper68_Asymmetry.lean`	183	Target 2: asymmetry theorem
Total	**493**	`sorry`: 0 warnings: 0 errors: 0

Build status: `lake build` → **0 errors, 0 warnings, 0 `sorry`s**. Lean 4 version: `v4.29.0-rc1`. Mathlib4 dependency via `lakefile.lean`.

5.2 Axiom inventory

The formalization declares axioms in two categories. First, 12 *opaque declarations* (`ArtinLocalRing`, `ArtinModule`, `embDim`, `IsFlat`, `IsFreeModule`, `NumberField`, `absDisc`, `ConjClass`, `frobInClass`, `ResidualRep`, `twSplittingField`, `TWConditions`) introduce the mathematical universe as uninterpreted types and properties. Second, 8 *theorem-level axioms* encode the deep results:

#	Axiom	Role	Reference
1	`brochard_finite_criterion` (B1)	Load-bearing	Brochard, *Comp. Math.* 153 (2017)
2	`effective_chebotarev` (B2)	Load-bearing	LMO (1979); Ahn–Kwon (2019)
3	`tw_disc_computable` (B3)	Documentation	Standard ANT (Hensel bounds)
4	`twConjClass`	Bridge	Standard ANT
5	`frob_implies_tw_conditions`	Bridge	Defn. of TW primes
6	`construct_patching_data`	Bridge	Wiles (1995), Diamond (1997)
7	`patching_data_valid`	Bridge	Wiles (1995), Diamond (1997)
8	`patching_data_edim`	Bridge	Wiles (1995), numerical criterion

Note on B3. Within the opaque-type framework, bare existence ($\exists d$, `absDisc` $L = d$) would be vacuously true. Axiom B3 strengthens this to $d > 0$, recording the mathematical content that the splitting field is a genuine number field. The primary role of B3 is as a *documentation marker* in the proof pipeline, flagging the discriminant computation as the input to the Chebotarev bound.

5.3 Key code snippets

CRM hierarchy (from `Paper68_Axioms.lean`):

```
inductive CRMLevel where
  | BISH | MP | LLPO | WLPO | LPO | CLASS
  deriving DecidableEq, Repr

def CRMLevel.join : CRMLevel -> CRMLevel -> CRMLevel
  | BISH,  b     => b
  | a,     BISH  => a
  | CLASS, _     => CLASS
  | _,     CLASS => CLASS
  | LPO,   _     => LPO
  | _,     LPO   => LPO
```

```
  | WLPO,   _      => WLPO
  | _,      WLPO   => WLPO
  | LLPO,   _      => LLPO
  | _,      LLPO   => LLPO
  | MP,     MP     => MP
```

TW prime search terminates (from `Paper68_Stage5.lean`):

```
theorem tw_prime_search_terminates
  (N p : Nat) (rhobar : ResidualRep N p) :
  exists (bound : Nat) (q : Nat),
    Nat.Prime q /\ q <= bound
    /\ TWConditions N p 2 q rhobar := by
  -- B3: compute discriminant (positive)
  obtain <d_L, _hpos, hdisc> := tw_disc_computable N p rhobar
  -- B2: effective Chebotarev
  obtain <q, hprime, hbound, hfrob> :=
    effective_chebotarev _ (twConjClass N p rhobar) d_L hdisc
  -- Frobenius => TW conditions
  exact <d_L ^ 12577, q, hprime, hbound,
    frob_implies_tw_conditions N p q rhobar hprime hfrob>
```

Asymmetry theorem (from `Paper68_Asymmetry.lean`):

```
theorem asymmetry_theorem :
    wiles_overall = CRMLevel.WLPO /\
    wiles_without_stage1 = CRMLevel.BISH :=
  <wiles_proof_classification, wlpo_localisation>
```

5.4 `#print axioms` output

Theorem	Axioms (custom only)
`tw_prime_search_terminates`	B2, B3, axioms 4–5
`stage5_is_bish`	B1, bridge axioms (6–8)
`wiles_proof_classification`	**None** (definitional `simp`)
`wlpo_localisation`	**None** (definitional `simp`)
`asymmetry_theorem`	**None** (pair of above two)

Classical.choice audit. The CRM hierarchy (`CRMLevel`) and the asymmetry theorem are purely inductive-type computations: `#print axioms asymmetry_theorem` shows only `propext` and `Quot.sound`. The Stage 5 theorems carry `Classical.choice` via the Mathlib `Nat.Prime` infrastructure—an artifact, not a proof-content dependency.

5.5 Reproducibility

Lean 4 formalization files are available at the Zenodo repository: https://doi.org/10.5281/zenodo.18748460. The bundle compiles with `lake build` on Lean v4.29.0-rc1 + Mathlib4.

6 Discussion

6.1 The de-omniscientizing descent pattern

Paper 50 [22] identified a "de-omniscientizing descent" in the five great conjectures: geometric origin converts LPO-level data to BISH-level data. Paper 68 reveals a parallel phenomenon in proof methods.

The descent $\mathsf{MP}+\mathsf{FT} \to \mathsf{BISH}$ occurred over twenty-two years (1995–2017), driven by algebraists solving commutative algebra problems, not by logicians. This is the first example in the CRM program of a de-omniscientizing descent occurring on a *human* timescale in the published literature, as opposed to a conjectural timescale.

6.2 What the calibration reveals

The deepest implication is structural: the Langlands correspondence, at least for $\mathrm{GL}_2/\mathbb{Q}$, has a logical asymmetry. The Galois side (deformation theory, patching) is constructive. The automorphic side (trace formula, L-functions) is not. The bridge between them adds no cost; the non-constructive content lives on the automorphic bank.

6.3 Relationship to existing literature

McLarty [15] showed Grothendieck universes are eliminable from FLT. Our analysis is orthogonal: set-theoretic strength vs. constructive principles. Mines–Richman–Ruitenburg [14] developed constructive commutative algebra, providing the framework for the BISH claims in Stages 2–4. The Taylor–Wiles method has been extended to GL_n by Barnet-Lamb–Gee–Geraghty [2] and to non-self-dual representations by Calegari–Geraghty [7]; the constructive status of these generalizations remains open.

6.4 What the Lean verification adds

The Lean 4 formalization verifies the *logical assembly*: given axiomatized inputs (Brochard, effective Chebotarev, the Taylor–Wiles construction), the composition yields the claimed CRM classifications. The axioms encode *precisely* where human mathematical judgment enters; the machine checks that no additional judgments are smuggled in.

This is the standard methodology for CRM formalization (cf. Paper 10 [20]). Formalizing Brochard's theorem or the Langlands–Tunnell theorem in Lean would be a multi-year project far beyond the scope of a single CRM audit. What the formalization achieves is a *machine-checked proof outline*: the deep mathematics lives in the axioms; Lean verifies that they compose correctly to produce the asymmetry theorem. The zero-`sorry` guarantee ensures no logical step has been skipped.

6.5 Is the WLPO intrinsic to FLT?

Fermat's Last Theorem is a statement about natural numbers: $\forall n \geq 3,\ \forall a\, b\, c \in \mathbb{N}^+,\ a^n + b^n \neq c^n$. As a Π^0_1 sentence, it "should" be provable in BISH if true. The WLPO in Wiles's proof is a feature of the *proof method*, not necessarily of the *theorem*.

This raises a fundamental question for CRM: does every true Π^0_1 sentence have a BISH proof? If the WLPO in Stage 1 is eliminable (e.g., via algebraic weight 1 modularity), this supports the CRM thesis that logical cost is intrinsic to theorems. If no proof of FLT avoids WLPO, this would be a striking counterexample: a concrete arithmetic statement whose *every* proof requires an omniscience principle.

The algebraic route via overconvergent p-adic methods (Buzzard–Taylor [6]) offers the most promising path to eliminating Stage 1's WLPO.

6.6 Open questions

1. **Is Stage 1 improvable?** Algebraic weight 1 modularity via overconvergent p-adic methods could drop Stage 1 to BISH. A constructive trace formula exploiting the effective Ramanujan bound might reduce WLPO to MP.
2. **Higher-dimensional modularity lifting.** Does the BISH classification of patching survive for GL_n? Barnet-Lamb–Gee–Geraghty [2] extend the method to GL_n; Calegari–Geraghty [7] handle non-self-dual representations. If patching remains BISH in these settings, the entire Langlands program's non-constructive content would be concentrated in the automorphic input.
3. **Constructive Euler systems.** Is the BISH classification of Rubin's Euler system (Stage 4 CM base case) robust? A detailed formal audit of the norm-compatibility relations and Kolyvagin's descent would strengthen the classification.

7 Conclusion

We have applied constructive reverse mathematics to Wiles's proof of Galois modularity for semistable elliptic curves over $\mathbb{Q}$ and hence Fermat's Last Theorem, and established that:

- The Taylor–Wiles patching step (Stage 5) is BISH, via Brochard's finite-level criterion and effective Chebotarev (Lean-verified from axioms).
- Stages 2–4 (deformation ring, Hecke algebra, numerical criterion, CM base case) are BISH (rigorous analysis, supported by Paper 59).
- Stage 1 (Langlands–Tunnell) requires at least WLPO (rigorous analysis of the trace formula and converse theorem).
- The overall classification is BISH + WLPO, with the WLPO localized entirely in Stage 1 (Lean-verified, sorry-free).

The constructive calibration does not resolve whether the WLPO is intrinsic to FLT, but it localizes the non-constructive content precisely: the Taylor–Wiles engine contributes zero logical cost; all omniscience lives in the harmonic analysis that provides the starting point. Geometric modularity ($E \sim A_f$, requiring Faltings) remains CLASS, but FLT needs only Galois modularity ($\rho_{E,p} \cong \rho_{f,p}$), which this paper classifies at BISH + WLPO. Whether this asymmetry is eliminable depends on a question at the frontier of the Langlands program: is there a purely algebraic proof of weight 1 modularity for solvable Artin representations?

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The deep mathematics is due to Wiles, Taylor, Diamond, Langlands, Tunnell, Brochard, and Lagarias–Odlyzko. The CRM methodology follows Bishop [3] and Bridges–Richman [4]. The Lean 4 formalization was produced using AI code generation (Claude Code, Opus 4.6) under human direction. The author is a practicing cardiologist rather than a professional logician or arithmetic geometer; all mathematical claims should be evaluated on their formal content. We welcome constructive feedback from domain experts.

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